

POLITECNICO
MILANO 1863

SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE

Executive Summary of the Thesis

Towards fast and side-channel resistant implementations of Hamming Quasi-Cyclic: a case study on ARM Cortex-M4 devices

Laurea Magistrale in Computer Engineering - Ingegneria Informatica

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Academic year: 2022-2023

1. Introduction

The advancements in quantum computing threaten the *status quo* in the cryptography field: a quantum computer with a sufficient amount of qubits is able, thanks to Shor's algorithm, to break most of the current public key cryptosystems in polynomial time.
For this reason in 2016 the National Institute of Standards and Technologies (NIST) launched a competition with the aim of developing and standardizing one or more quantum-resistant cryptosystems. After three rounds CRYSTALS-Kyber, a lattice-based cryptosystem, was selected as standardization candidate; however, in the event of a potentially disruptive attack on lattice-based cryptosystems, NIST decided to further extend the competition selecting four more candidates that rely on different hard problems. Excluding SIKE, the remaining candidates, namely BIKE [1], HQC [5] and Classic McEliece [2] are all based on the hardness of the syndrome decoding problem.
The object of this work is HQC, a code-based cryptosystem whose security relies on the syndrome decoding problem for quasi-cyclic codes. HQC comprises both a Public Key Encryption (PKE) scheme and a Key Encapsulation Mechanism (KEM), obtained by applying the HHK Fujisaki-Okamoto transformation on the former.

Related work Given the sound security record of HQC, much attention has been devoted to its side-channel resistance. Most of the independent analyses focus on extracting information from electromagnetic and power traces, whilst at the end of Round 2 of the NIST contest the authors in [7] and [6] proposed a timing attack that takes advantage of the non constant-time implementation of the BCH decoder; these attacks are no longer relevant, as the authors of HQC swapped the BCH code with a Reed-Muller Reed-Solomon concatenated code, implementing a constant-time decoder. Nontheless, in [3] the authors propose a new timing attack, based on a weakness in the *rejection sampling* procedure, employed to generate random sparse polynomials; as a countermeasure, a more care careful implementation of the rejection sampling procedure has been implemented in Round 4, fixing the issue.
To date, no dedicated constant-time implementation of HQC targeting embedded systems has been proposed, despite the fact that side channel attacks are easier to carry on such platforms.

The current state of software implementations of HQC tailored for the ARM Cortex-M4 microcontroller, chosen by the NIST as the reference platform for microcontroller benchmarks, includes only the one in PQM4 [4], a third-party public project that collects the implementations of NIST Round 4 candidates tailored for Cortex-M4. The latest version of HQC in PQM4 is directly derived from the Round 3 reference implementation, with no optimizations: as a consequence, this implementation is not constant-time and vulnerable to the timing attack presented in [3].

Our contribution In this work we present the first fully constant-time software implementation of HQC, together with a comparative analysis of constant-time binary polynomial multiplication techniques, both tailored for ARM Cortex-M4 microprocessors. We provide an experimental validation of the constant-time properties via a battery of statistical t-tests, while varying the CPU frequency to highlight the stability of our solution across different memory access latencies. Our results show that the optimal solution for the target platform is a FFT-based approach, in contrast with classic divide-et-impera techniques (*i.e.* Karatsuba) which provide better performance on x86 architectures.
Moreover, to counter power attacks, we propose a masking scheme that can be applied to the entire cryptosystem, presenting our results on its potential efficacy. Finally, we compare our work with the latest implementations available of BIKE and Classic McEliece, showing the high efficiency of our approaches.

2. Background

3. Implementation

4. Experimental results

5. Conclusions

A final section containing the main conclusions of your research/study have to be inserted here.

References

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